ARLO INDUSTRIES

PASSIVE AERIAL SENSING MESH 15X BETTER THAN RADARS

Deep Tech / Dual-Use / Space

TEAM

DEO ARLO
CEO

2x Founder with 6 years of experience in the defence ecosystem in Israel. Robotics and AI researcher at Technion (Patented Inventor). Born during riots, defence is his life mission.

JAINIK MEHTA
ENGINEERING

PhD in Mechatronics and Robotics at Rensselaer Polytechnic. Worked on UAS, LiDAR, and DDIL in IIT Bombay and Technion. Expert in sensor fusion, perception algorithms, and signal processing.

DEFENCE CONFERENCE - TEL AVIV

MENTORED BY ALEX BOUAZIZ

IGSTC - DARMSTADT GERMANY

THE RESIDENCY - SF

HISTORY

Deo and Jainik have known each other for 6+ years, lived as flatmates at Technion, and won competitions together.

2019 - TECHNION

2026 - Y COMBINATOR

FOUNDERS INC - SF

FR8 - FINLAND

MENTORS

ANTTI KOSUNEN — INDUSTRY
DEFENCE ECOSYSTEMS AND MARKET ACCESS

Accelerator builder with extensive experience connecting defence startups with operators, investors, and deployment partners across Europe and Ukraine.

MARTTI WALLIN - COMPLIANCE
EXPORT CONTROLS & EU DEFENCE

Finnish defence and aerospace advisor. Industrial Counsellor at the Embassy of Finland in Washington DC. Longtime executive and export-controlled environments.

SAMI LUUKKONEN — STRATEGY AND GROWTH
DEFENCE, AEROSPACE AND GLOBAL EXPANSION

Global technology executive and investor. Managing Director at Accenture and senior executive at Singtel NCS, with decades of experience in aerospace and defence.

SUPPORT

INSTITUTIONS

TECHNION Israel Institute of Technology

Rensselaer

PROGRAMS

Y Combinator

17 Tech

TESTING PARTNERS

JIFX Joint Interagency Field Experimentation

defence builder

17 TECH DEFENCE ACCELERATOR - **FINLAND**

DRAGON SKY & DEFENCE BUILDER - **UKRAINE**

AERIAL DEFENSE IS BROKEN

- US is building **"Golden Dome" (2028)** & NATO **"Drone Wall" (EDDI 2027)**
- **Drone warfare spilling over into EU:** confirmed incursions over Poland, Estonia, Denmark, Germany, and Finland.
- There are tons of interceptors but very little innovation in sensing architecture. **Radars are a 100+ year old technology.**
- DoD LIDS architecture & EU C-UAS gaps: both rely on **high-cost** radar.
- **3838 Act** mandates edge AI in defence procurement.
- **C-UAS** across all domain is a **$440B Market**

ROCKET PASSES IRON DOME

ROCKETS AROUND MY APARTMENT IN TEL AVIV

NBC NEWS A 'drone wall'? Europe floats a high-tech SHARE & SAVE

Instead, Tollast described a potential network of small, mobile tactical radars and sensors to spot low-flying drones, linked to mobile air-defense teams and fixed batteries that use rapid cannon fire. Higher-altitude targets would require their own defenses.

We are building exactly this

the summits, after the
ntinent in recent

America's 'Golden Dome' Explained

SON/AFP/Getty Images

WAR IN UKRAINE

A 'drone wall'? Europe floats a high-tech response to Russian threats

Europe's leaders gathered in Copenhagen this week after a wave of drone incursions above airports and sensitive sites across the continent.

PROBLEM

Shaheds and FPV drones fly slow & low below radar horizons and often **undetectable** amid clutter.

Traditional RF/mmWave systems struggle to identify them, especially in urban terrain.

Systems like Robin Radar cost **$500k** **per unit** with only 12 km range. US army is paying KuRFS + Coyote Radars for **$5.04B** through 2033.

Ukrainian analysis on frontline sensing gaps

Acoustic meshes allow large coverage but **low fidelity** while RF radars offer high fidelity but are scarce and **doesn't scale** effectively.

CSIS report on distributed Mesh Sensing

SOLUTION

The world's first Optical Mesh *Radar*. Each node is deployable in ~1 minute by a single operator.

Range 3x Radars

Future versions can even track meteorites & Stealth Aircrafts (50-60km) Competitors only 4-15km.

* Theoretical estimation, technical validation will come with benchmarking in coming months

Affordable Full Coverage

Covers Kyiv sized area **~1000 km² for <$20M**, while competitors $30M-300M with less range.

Scalable

With each node the performance performance grows exponentially, while cost linearly.

Market Size

TAM $440B - C4ISR Sensing + Border + Critical Infrastructure + Airport

SAM $60B - US, NATO + allied nations

SOM $400M - 200 deployments by year 5 with avg deal: **$1-3M**

Go-To-Market Strategy

- Critical Infrastructure: Power plants, Airports
- C-UAS Interceptors - help guide interceptors
- Radar companies - augment existing sensing

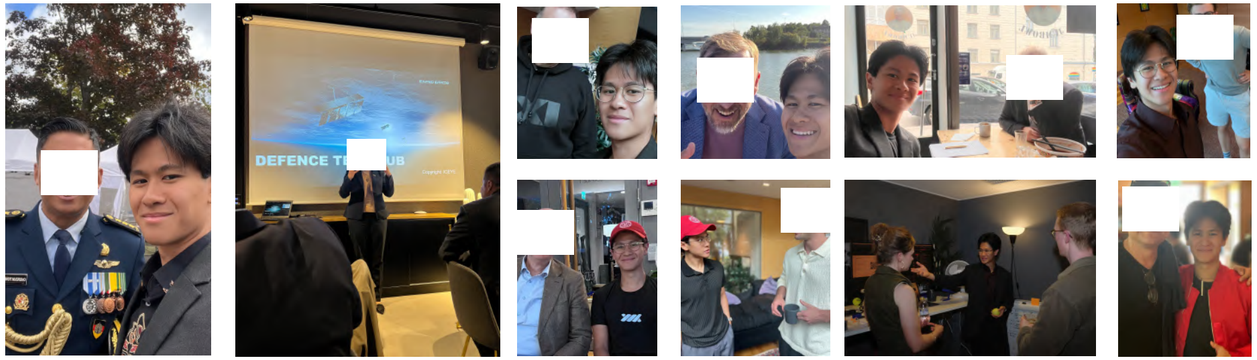
Current stage:

Mentat is currently at **TRL-5**. A multi-node prototype has been built and demonstrated. The core voxel-based sensing algorithm has been validated on real aerial targets under controlled conditions.

Field testing with defence partners is underway to transition the system to **TRL-7** with testing in non-ideal conditions and increased range. Pilot underway in Estonia, testing in Finland and Ukraine.

WE ARE DEAD SERIOUS

We have done **hundreds** of both in-person and online research with: Defence Companies, Industry experts, and Operators from USA, Ukraine, Germany, France, Finland, Estonia, Lithuania, Latvia, Poland, and more



faces blurred to protect their privacy

PROJECT TIMELINE – 2026

Q1: Field Testing with Defence Partners

- Live system testing with Finnish and Ukrainian defence partners
- Collect real-world performance data on detection range, false positives, deployment speed, and robustness
- Validate multi-node coordination under operational conditions

Q2: Iteration & US DoD Benchmark Preparation

- Incorporate feedback and improve system from Q1 testing
- Participate in US Department of Defense JIFX testing and benchmarking

Q3: Pilot Deployments

- Deploy Mentat in pilot configurations with defence and dual-use partners in US & EU
- Validate operational workflows, installation procedures, and sustainment requirements
- Gather operator feedback and pilot-level performance metrics

Q4: Post-Benchmark Improvement & Critical Infrastructure

- Improve system performance based on DoD benchmark results
- Harden hardware and software for reliability and repeatability
- Deploy Mentat for initial critical infrastructure protection use cases

THE FUTURE

Although our GTM is defence, the long-term vision is a planetary and orbital sensing architecture. My earlier work proved that distributed, low-power nodes can turn even the most remote environments into **self-aware digital systems**. I deployed an edge-compute and Starlink-connected monitoring network in the Congo rainforest with conservation groups and Microsoft, showing that **large-scale, resilient sensing** grids can operate where traditional infrastructure cannot.

Mentat extends that philosophy from the ground into the **stratosphere and orbit**. As the mesh matures, it becomes part of a multi-layered Earth-space system for tracking aircraft, drones, satellites, and space debris in real time. The same passive localisation stack we use today for aerial threats can scale into **Orbital Domain Awareness**, one of the most critical emerging challenges as LEO becomes congested with proliferated constellations.

Beyond Defence:
We have secured future space partner companies

- Space debris and satellite tracking across LEO; a passive, distributed layer to augment existing radar/telescope systems.
- Micro-satellite constellation for Earth-focused sensing and orbital monitoring.
- Orbital Domain Awareness for manoeuvre detection, proximity events, and uncooperative objects.
- Deep-space monitoring for NEOs, cislunar traffic, and interplanetary missions.

LET'S TALK

DEO@ARL01.COM